

FOUNDATION

Why Audio Fundamentals Matter



Voice AI systems don't understand speech directly—they interpret numerical representations of audio. The quality of these representations fundamentally shapes system performance.

Audio quality directly influences:

- Automatic speech recognition (ASR) accuracy
- System latency and response time
- Storage requirements and costs
- Streaming performance
- Overall user experience

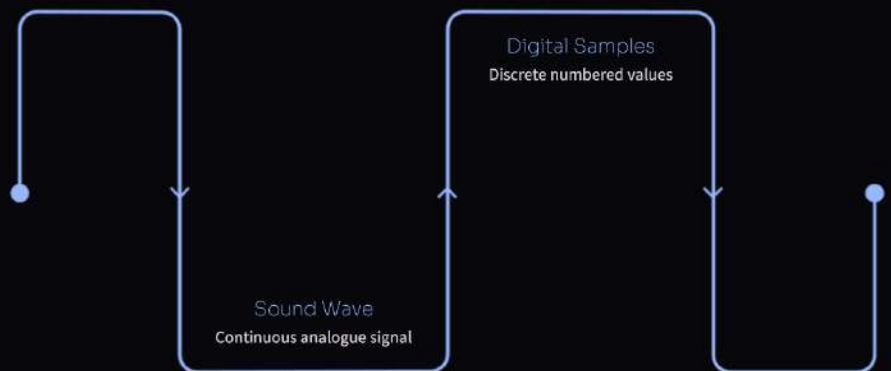
What Is Audio from a Computer's Perspective?

The transformation

Sound exists as a continuous physical wave in nature. Computers must convert this infinite analogue signal into discrete numerical samples to process it.

Digital audio = a sequence of numbers representing sound amplitude measured at specific points in time.

Each sample captures the instantaneous amplitude of the sound wave, creating a numerical fingerprint of audio.



Analogue vs Digital Audio

Analogue Audio

- Continuous waveform** with infinite resolution
- Exists in the physical world as pressure variations
- Can't be directly processed by computers

Digital Audio

- Discrete samples** stored as numerical values
- Computer-readable format enabling AI processing
- Required for all Voice AI systems

Voice AI exclusively operates on digital audio, making the conversion from analogue to digital a critical first step in any speech recognition pipeline.

What Is Sampling?

Sampling is the process of converting continuous sound into discrete digital measurements that computers can process.

01

Capture

Microphone detects sound pressure variations in the air

02

Measure

System records amplitude values at fixed time intervals

03

Store

Measurements become samples forming digital audio data



What Is Sample Rate?

Definition

Sample rate measures how many amplitude samples are captured per second of audio.

Unit: Hertz (Hz)

Higher sample rates capture more detail but create larger files.

Common sample rates

- 8,000 Hz → 8,000 samples per second
- 16,000 Hz → 16,000 samples per second
- 44,100 Hz → 44,100 samples per second
- 48,000 Hz → 48,000 samples per second

The Nyquist theorem states that sample rate must be at least twice the highest frequency you want to capture.

Sample Rate Visualisation

The trade-off is clear: higher sample rates preserve more acoustic detail and improve speech clarity, but they significantly increase file size and processing requirements. Finding the optimal balance is essential for Voice AI systems.

INDUSTRY STANDARDS

Common Sample Rates in Voice AI



8 kHz

Telephony standard

Traditional phone calls and low-quality speech applications



16 kHz

ASR standard

Optimal balance of quality and efficiency for speech recognition systems



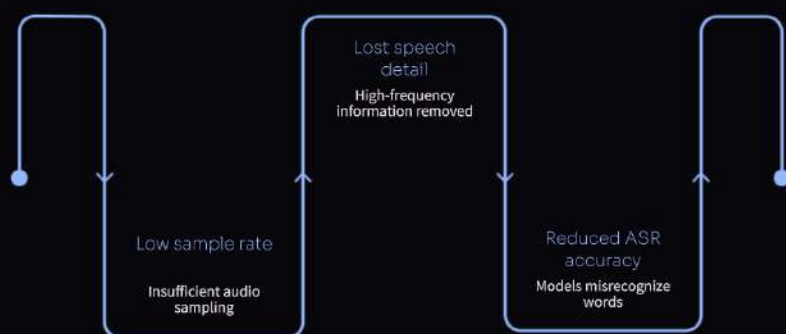
44.1 kHz

Audio CD quality

Music production and high-fidelity recordings requiring full frequency range

❑ Most ASR models are trained and optimised for 16 kHz audio, making it the industry standard for Voice AI applications.

What Happens When Sample Rate Is Too Low?



Consequences of insufficient sampling

When sample rate falls below optimal thresholds, critical speech information disappears, creating a cascade of problems:

- **Aliasing**—high-frequency components misrepresented
- **Missing phonetic details**—subtle speech sounds lost
- **Reduced intelligibility**—harder for models to distinguish words
- **Higher error rates**—transcription accuracy degrades measurably

Testing shows ASR accuracy can drop 15-30% when moving from 16 kHz to 8 kHz audio.

What Is Bit Depth?

Definition

Bit depth determines how many bits represent each audio sample, directly affecting dynamic range and precision.

Common Values

- **8-bit**—256 possible values
- **16-bit**—65,536 values (standard)
- **24-bit**—16.7 million values

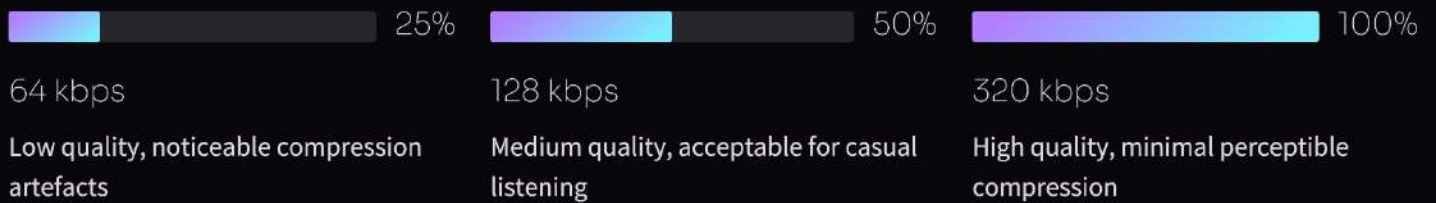
Impact

Higher bit depth provides better dynamic range, capturing both quiet and loud sounds with greater precision.

Most ASR systems use 16-bit audio as it provides excellent quality whilst maintaining reasonable file sizes. The 96 dB dynamic range of 16-bit audio is more than sufficient for speech applications.

What Is Bitrate?

Bitrate measures the amount of data used per second of audio, combining sample rate, bit depth, and compression into a single metric.



Sample Rate vs Bitrate

Sample Rate

What it measures: Frequency of sampling (samples per second)

What it affects: Frequency response and temporal resolution

Example: 16,000 Hz captures 16,000 amplitude measurements every second

Bitrate

What it measures: Data throughput (bits per second)

What it affects: Overall quality, file size, and streaming requirements

Example: 256 kbps uses 256 kilobits of data per second

Both parameters significantly influence quality, storage requirements, and transmission speed. Understanding their relationship helps optimise Voice AI deployments.

What Are Audio Channels?

Mono

Single channel containing one stream of audio data

Standard for ASR systems—more efficient processing

Stereo

Two independent channels (left and right) creating spatial sound

Used for music and immersive audio experiences

- ❑ Voice AI systems typically use mono audio because speech recognition doesn't benefit from stereo separation. Mono also halves file size and processing requirements compared to stereo.

What Is an Audio Format?

Audio formats define the file structure and encoding method used to store digital audio data. The choice of format significantly impacts quality, compatibility, and processing efficiency.



Container formats

Define file structure: WAV, MP3, M4A, FLAC, OGG



Encoding methods

Compression algorithms: PCM, MPEG, AAC, Opus



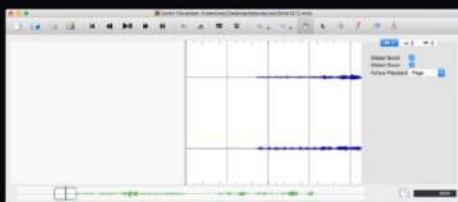
Key factors

Quality preservation, file size, processing speed, compatibility



★ RECOMMENDED

WAV Format



Lossless uncompressed audio

WAV (Waveform Audio File Format) stores raw audio data without any compression, preserving complete audio fidelity.

Key characteristics

- Zero compression—no data loss whatsoever
- Maximum quality—perfect reproduction of original
- Large file size—approximately 10 MB per minute (16-bit, 16 kHz, mono)
- Universal compatibility—supported by virtually all systems

WAV is the preferred format for Voice AI applications and model training.

MP3 Format

Characteristics

MP3 uses lossy compression based on psychoacoustic models—it removes audio information that humans theoretically can't perceive.

- **Compression ratio:** 10:1 to 12:1 typical
- **File size:** ~1 MB per minute (128 kbps)
- **Quality loss:** Irreversible data removal
- **Compatibility:** Universal playback support

Not ideal for ASR

Whilst MP3 works for human listening, the compression removes frequencies and subtle details that ASR models use for accurate transcription.

Testing shows MP3 encoding can reduce ASR accuracy by 5-15% compared to WAV, particularly for challenging audio conditions.

Lossless vs Lossy Formats

The fundamental trade-off

Lossless formats preserve every bit of audio data, ensuring perfect reproduction. Lossy formats sacrifice some data to dramatically reduce file size.

Voice AI requirements

ASR systems benefit significantly from lossless formats because they preserve the acoustic features models rely on for accurate speech recognition.

Why Audio Format Affects ASR Accuracy



Compression

Missing Data

Lower
Confidence

Degraded
Accuracy



BEST PRACTICES

Recommended Audio Settings for Voice AI



Format

WAV (PCM encoded)

Lossless quality, universal compatibility



Sample Rate

16 kHz

Optimal for speech, efficient processing



Channels

Mono

Single channel sufficient for speech



Bit Depth

16-bit

Excellent dynamic range, standard size

This configuration delivers the optimal balance of audio quality, ASR accuracy, file size, and processing efficiency for production Voice AI systems.